

UMER AKBAR

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PROFILE

Future-focused AI engineering student (7th semester) specializing in Generative AI, Agentic AI, and Computer Vision. Proficient in designing robust ML pipelines and real-world AI systems using Python, PyTorch, and TensorFlow. Seeking an internship or entry-level role to leverage hands-on expertise in intelligent automation, scalable AI architectures, and MLOps methodologies.

PROFESSIONAL EXPERIENCE

- Agentic AI Engineer Intern** | Inventacore 08/2026 – Present | Islamabad, Pakistan
- Engineered and deployed a live cloud-hosted automated compliance engine featuring video walkthrough documentation and a public repository.
- Generative AI Engineer** | Decodelabs Present | Islamabad, Pakistan
- Engineered and deployed cutting-edge Generative AI models and integrated LLM solutions.
- Machine Learning Intern** | Digital Empowerment Network 07/2025 – 09/2025 | Islamabad, Pakistan
- Developed a Text Classification model with NLP and word embeddings, a Fake News Detector, and a Multimodal Housing Price Prediction system.
- Data Science Intern** | OASIS INFOBYTE 06/2025 – 07/2025 | Islamabad, Pakistan
- Implemented clustering and dimensionality reduction techniques for complex datasets.
- AI & ML Engineering Intern** | Developers HUB Corporation 06/2025 – 07/2025 | Islamabad, Pakistan
- Engineered ML pipelines including the deployment of a sentiment analysis model.
- Machine Learning Intern** | Certura 05/2025 – 06/2025 | Islamabad, Pakistan
- Contributed to machine learning projects and pipelines.
- Machine Learning Intern** | CodeAlpha Islamabad, Pakistan
- Designed predictive models encompassing handwritten character recognition, credit scoring evaluation, and disease prediction pipelines.

PROJECTS

Automated Code-Review Program

- Engineered a full-stack automated code-review program utilizing a FastAPI server framework and an asymmetric frontend interface layout.

Automated Compliance Engine / Inventacore

- Engineered and deployed a live cloud-hosted automated compliance engine featuring video walkthrough documentation and a public repository.

Generative AI Solutions / Decodelabs

- Engineered and deployed cutting-edge Generative AI models and integrated LLM solutions.

Agentic Employment Law Assistant / Source-Grounded Legal RAG Architecture

- Architected a secure, local Retrieval-Augmented Generation (RAG) system using LangChain, FAISS, and Hugging Face embeddings to parse complex regulatory frameworks with zero hallucination.
- Engineered a dynamic document ingestion pipeline and Streamlit dashboard that automatically generates deterministic, verifiable source citations (document, page, and jurisdiction) for strict legal compliance.

Agentic E-Commerce System / Autonomous DM Agent & Market Intelligence

- Engineered a dual-agent AI suite for high-volume Shopify storefronts to automate competitive analysis and customer interactions.
- Developed a B2B intelligence agent using CrewAI to simulate competitor data scraping and generate actionable weekly merchandising reports via LLMs.
- Built a B2C customer agent with a Streamlit frontend employing function-calling to autonomously handle queries, check live inventory, and generate secure checkout links.

Self-Correcting RAG / *Corrective RAG Architecture*

- Developed an advanced RAG system using PyTorch, FAISS, LangChain, and FastAPI to process technical documentation.
- Integrated a local AI evaluator (Ollama + Mistral 7B) to judge retrieved document relevance before generating responses.
- Implemented a self-correcting fallback mechanism that autonomously searches the web (Tavily API & Groq API) when local documents fail the quality threshold.

Agentic Inspector / *Real-Time Quality Assurance System*

- Integrated a ResNet18 architecture with the Gemini API to analyze visual data and deliver immediate, intelligent quality assessments.
- Implemented Explainable AI (XAI) using Chain-of-Thought (COT) prompting for transparent model reasoning, deployed via Streamlit.

AI Chatbot / *Conversational AI Interface*

- Engineered a custom conversational agent leveraging large language models and natural language processing to handle dynamic, multi-turn user interactions.
- Developed a responsive frontend using Streamlit and Python, incorporating contextual memory to maintain accurate and coherent dialogue histories.

SKILLS

Languages:	Python, C++, SQL
ML Frameworks:	PyTorch, TensorFlow
Core Domains:	Computer Vision, NLP, Generative AI
Agentic AI:	LangChain, FAISS, CrewAI
Data Science:	Scikit-Learn, Pandas, NumPy, EDA
Tools:	Streamlit, FastAPI, Git, MySQL

EDUCATION

Bachelor of Science in Artificial Intelligence (7th Semester)	09/2023 – 05/2027
Shaheed Zulfiqar Ali Bhutto Institute of Science and Technology (SZABIST)	Islamabad, Pakistan